

Employee Payslip Generator

An automated software system that streamlines payroll processing — calculating net salaries, generating payslips, and reducing manual errors for organizations of all sizes.

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Why Automate Payroll?

Error-Prone Manual Methods

Spreadsheets and manual calculations lead to mistakes, employee dissatisfaction, and loss of trust.

Time-Consuming Processes

Processing large workforces manually can take hours or days — automation completes it in seconds.

Record Management Challenges

Manual records risk data loss and mismanagement, especially during audits or financial reviews.

Key Benefits

Time Efficiency

Automated salary calculations and payslip generation save hours of HR effort every pay cycle.

Accuracy

Predefined formulas eliminate human error, ensuring consistent and precise salary outputs.

Security

Authentication, role-based access, and data encryption protect sensitive payroll information.

Flexibility

Configurable salary structures and deduction rules adapt to any organization's policies.



Salary Calculation Formula

Earnings

Effective Basic = (Basic Salary ÷ 30) × Days Worked

HRA = 20% of Effective Basic

Medical Allowance = ₹ 1,500 (fixed)

Conveyance = ₹ 800 (fixed)

Gross Salary = Effective Basic + HRA + Medical +
Conveyance

Deductions

PF = 12% of Effective Basic

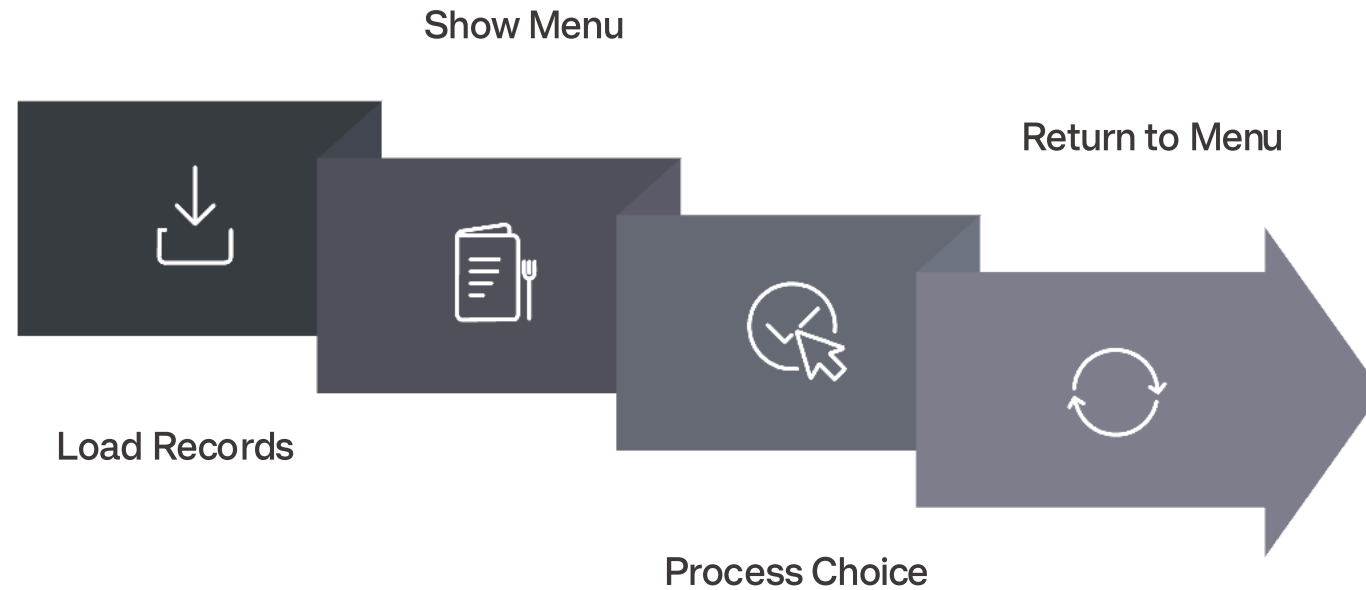
ESI = 1.75% of Gross Salary

Income Tax = 10% of Gross Salary

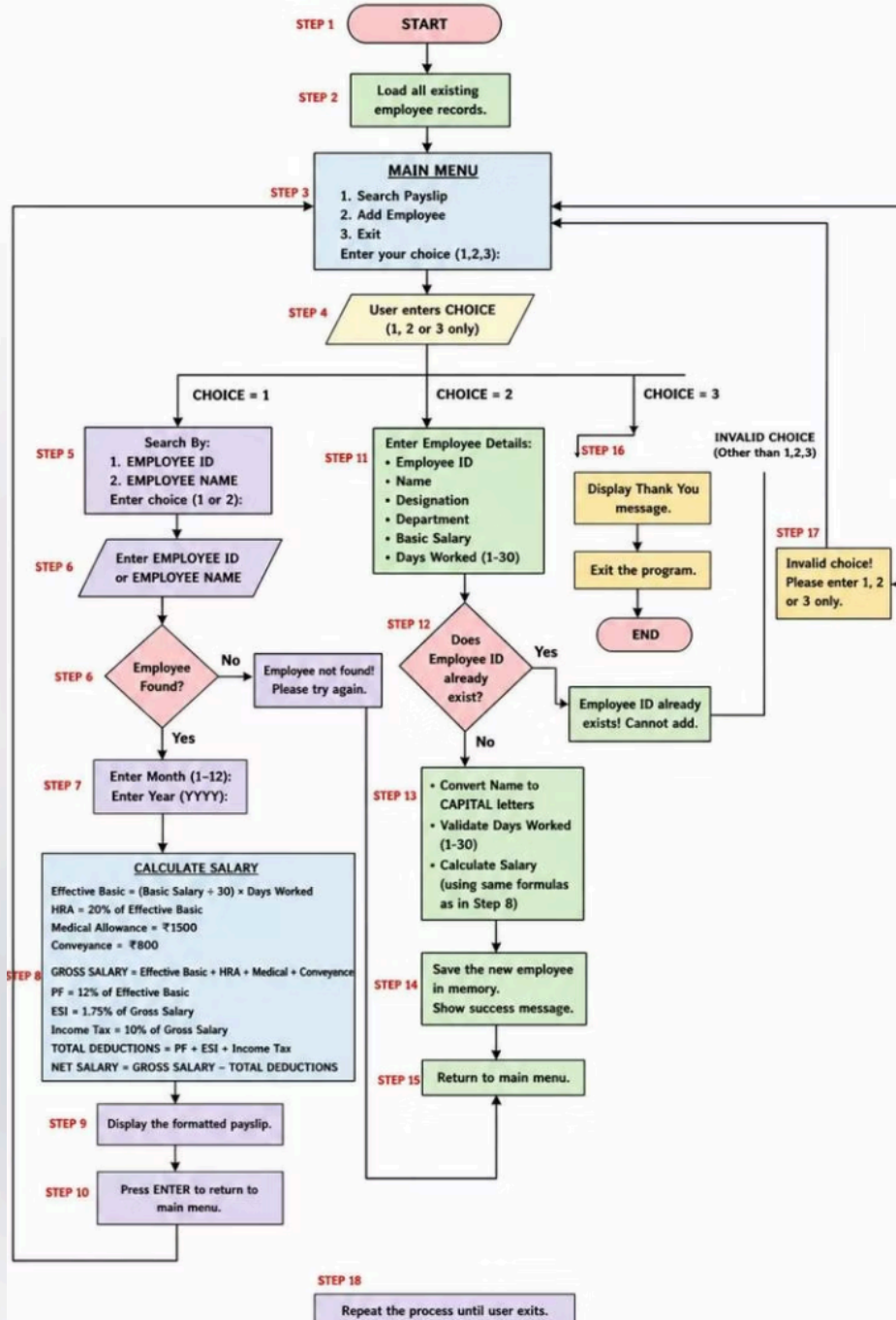
Total Deductions = PF + ESI + Income Tax

Net Salary = Gross Salary – Total Deductions

System Algorithm Overview



The system runs a continuous loop — loading records at startup, presenting a 3-option menu, processing each user choice, and returning to the menu until the user exits.



Program Flowchart

Key Design Principles

Data Persistence

Records loaded at startup; new entries saved to memory throughout the session.

Input Normalization

Employee names auto-converted to CAPS for consistent and reliable searching.

Defensive Validation

Handles duplicate IDs, invalid days worked (1- 30), and unrecognized menu choices gracefully.

Loop Continuity

System returns to the main menu after every operation until the user explicitly exits.

Sample Output — Payslip

EMPLOYEE PAYSIP

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Employee ID : 0439408

Name : KAVITA SINGH

Designation : MUSIC TEACHER

Department : ARTS & LANGUAGE

Month / Year : 6/2024

--- EARNINGS ---

Effective Basic 37000.00

HRA (20%) 7400.00

Medical Allowance 1500.00

Conveyance 800.00

GROSS SALARY 46700.00

--- DEDUCTIONS ---

PF (12%) 4440.00

ESI (1.75%) 317.25

Income Tax (10%) 4670.00

TOTAL DEDUCTIONS 9927.25

NET SALARY 36772.75

*** THANK YOU ***

How It Works

Search by Employee ID or Name, enter the month and year, and the system instantly calculates and displays a fully formatted payslip.



Employee POOJA TIWARI (EVS Teacher) — Net Salary:
₹ 39,589.75 for June 2025



New employee ANAND SHARMA added successfully with
Basic Salary ₹ 38,000 and 28 days worked.



Conclusion

The Employee Payslip Generator is an effective, reliable payroll solution that automates salary calculations, minimizes errors, and ensures transparency. Its user-friendly design, secure record management, and flexible configuration make it an essential tool for modern organizations of all sizes.

Future Enhancements

Cloud & Mobile

Cloud-based access and a dedicated mobile app with push notifications for new payslips.

Real-Time Tax Compliance

Auto-updated tax rules, multi-currency, and multi-language support for global organizations.

HRMS Integration

Seamless data sharing with HR management and recruitment software to reduce duplication.

AI & Analytics

Salary trend insights, compensation pattern analysis, and cost optimization via AI-driven dashboards.